



برنامج الأمن السيبراني
Cybersecurity Program

02203N : Cryptography

MidTerm Exam - March, 2026

25 MCQs

Spring 2025/2026

Duration : 60 Minutes

ID: _____ Name: _____

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Documents and Electronic Facilities (*calculators or smart phones*) are forbidden.

Question 1 Block ciphers accumulate symbols in a message of:

- ☐ A Variable size. ☐ B Integration size. ☒ C Fixed size. ☐ D Increasing size.

Question 2 What is the block size and key length for the IDEA Cryptography algorithm?

- ☐ A 64, 56 bits. ☒ B 64, 128 bits. ☐ C 128, 256 bits. ☐ D 64, 64 bits.

Question 3 Which of the following is / are the application(s) of the DES algorithm?

- ☐ A ATM. ☐ B E-mail. ☐ C Remote access. ☒ D All.

Question 4 The sub key length at each round of DES is:

- ☒ A 48 ☐ B 32 ☐ C 56 ☐ D 64

Question 5 Which of the following is the fourth step of AES?

- ☐ A Sub bytes. ☒ B Add round keys. ☐ C Shift rows. ☐ D Mix columns.

Question 6 Which of the below security encryption standards is the strongest?

- ☐ A WPA ☒ B WPA3 ☐ C WEP ☐ D WPA2

Question 7 In which type of cryptography, same key is used for encryption and decryption?

- ☒ A Symmetric. ☐ B Asymmetric. ☐ C Public. ☐ D All.

Question 8 Which principle of cryptography explains transmitted data in cannot be modified?

- ☐ A Confidentiality. ☒ B Integrity. ☐ C Authentication ☐ D None.

Question 9 What is the output of a cryptographic hash function means?

- ☐ A A variable set of bits. ☐ C A fixed set of bits, bidirectional mapped.
☒ B A fixed set of bits, one-way mapped. ☐ D None.

Question 10 Which security services cannot be achieved using Hash functions?

- ☒ A Original data retrieval. ☐ C Digital signatures.
☐ B Data integrity check. ☐ D Password check.

Question 11 All of the following are cryptographic applications, except for:

- ☐ A Data integrity. ☐ B Confidentiality. ☐ C Authentication ☒ D None.

Question 12 Which one is preferred as encryption key in asymmetric cryptography?

- ☒ A Public key. ☐ B Private key. ☐ C Shared key. ☐ D None.

- Question 13** Which ciphers can operate on 64 bit blocks of plain and cipher data.
☐ A IDEA ☐ B CAST ☐ C Triple-DES ☒ D All
- Question 14** Which of the following is not a hash function?
☐ A MD5 ☐ B SHA-1 ☐ C SHA-256 ☒ D Blowfish
- Question 15** Which of the following algorithm(s) has overcome the DES?
☐ A CAST. ☐ B AES. ☐ C 3DES. ☒ D All.
- Question 16** Which block cipher has key length of 128 bits?
☐ A IDEA ☐ B CAST ☒ C Both (A) & (B) ☐ D None
- Question 17** Which is the monoalphabetic cipher that uses modular arithmetic with value 26?
☐ A transposition. ☐ B Additive. ☒ C Shift. ☐ D None.
- Question 18** In cryptography, what is cipher?
☒ A An algorithm for encryption / decryption. ☐ C The decrypted message.
☐ B The encrypted message. ☐ D None.
- Question 19** What is the length of the Hash value output of the SHA-1 algorithm?
☐ A 128 bits. ☒ B 160 bits. ☐ C 256 bits. ☐ D 512 bits.
- Question 20** At which end of the communicating parties decryption is performed at?
☐ A Transmitter. ☒ B Receiver. ☐ C Both ends. ☐ D None.
- Question 21** How many rounds does 256 bits key in AES are required?
☐ A 16 ☐ B 12 ☐ C 10 ☒ D 14
- Question 22** In cryptography, the order of letters in a message is rearranged by which cipher?
☒ A transpositional. ☐ B substitution. ☐ C sequential. ☐ D None.
- Question 23** Which cipher has two categories: monoalphabetic and polyalphabetic?
☒ A Substitution. ☐ B Transposition. ☐ C Symmetric. ☐ D None.
- Question 24** Which of the following is not a part of the rounds in the AES?
☐ A Mix column. ☒ B IP and IP^{-1} .
☐ B Byte substitution. ☐ D Shift row and add round key.
- Question 25** Which of the following is / are offered by the Hash functions?
☐ A Authentication. ☐ B Non repudiation. ☐ C Data integrity. ☒ D All.

Answer sheet:

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Answers must be given here: answers on other sheets will be ignored.

- QUESTION 1 : ☐ A ☐ B ☒ C ☐ D
- QUESTION 2 : ☐ A ☒ B ☐ C ☐ D
- QUESTION 3 : ☐ A ☐ B ☐ C ☒ D
- QUESTION 4 : ☒ A ☐ B ☐ C ☐ D
- QUESTION 5 : ☐ A ☒ B ☐ C ☐ D
- QUESTION 6 : ☐ A ☒ B ☐ C ☐ D
- QUESTION 7 : ☒ A ☐ B ☐ C ☐ D
- QUESTION 8 : ☐ A ☒ B ☐ C ☐ D
- QUESTION 9 : ☐ A ☒ B ☐ C ☐ D
- QUESTION 10 : ☒ A ☐ B ☐ C ☐ D
- QUESTION 11 : ☐ A ☐ B ☐ C ☒ D
- QUESTION 12 : ☒ A ☐ B ☐ C ☐ D
- QUESTION 13 : ☐ A ☐ B ☐ C ☒ D
- QUESTION 14 : ☐ A ☐ B ☐ C ☒ D
- QUESTION 15 : ☐ A ☐ B ☐ C ☒ D
- QUESTION 16 : ☐ A ☐ B ☒ C ☐ D
- QUESTION 17 : ☐ A ☐ B ☒ C ☐ D
- QUESTION 18 : ☒ A ☐ B ☐ C ☐ D
- QUESTION 19 : ☐ A ☒ B ☐ C ☐ D
- QUESTION 20 : ☐ A ☒ B ☐ C ☐ D
- QUESTION 21 : ☐ A ☐ B ☐ C ☒ D
- QUESTION 22 : ☒ A ☐ B ☐ C ☐ D
- QUESTION 23 : ☒ A ☐ B ☐ C ☐ D
- QUESTION 24 : ☐ A ☐ B ☒ C ☐ D
- QUESTION 25 : ☐ A ☐ B ☐ C ☒ D